

PHYS 230 - Electricity and Magnetism

Lecture 30

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

7 Magnetic Field and Magnetic Forces

- Magnetic Force and Torque on a Current-carrying Conductor (Continued)

8 Sources of Magnetic Field

- Magnetic Field of a Moving Charge

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

Outline

7 Magnetic Field and Magnetic Forces

- Magnetic Force and Torque on a Current-carrying Conductor (Continued)

8 Sources of Magnetic Field

- Magnetic Field of a Moving Charge

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

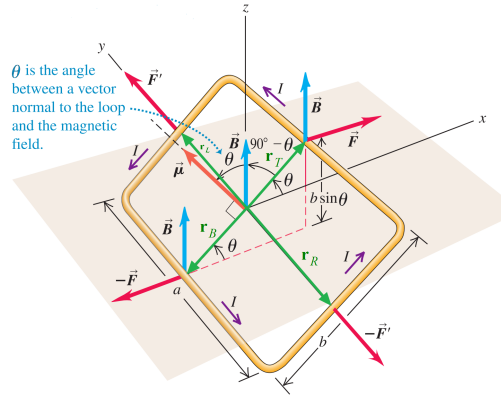
Torque on a Current Carrying Loop in a Magnetic Field

In this current-carrying loop, the torque on each side is

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$$

where $\mathbf{r}$ is the distance from the center of the loop to the center of that side, and $\mathbf{F}$ is the force on that side and the total torque on the loop is

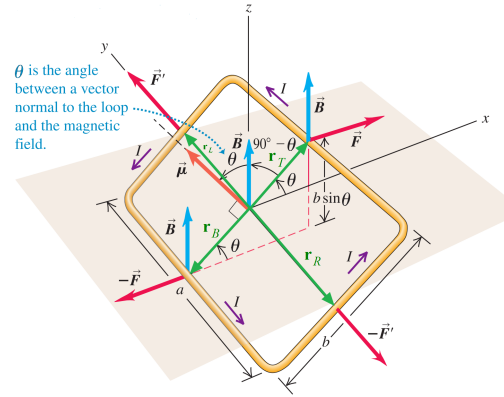
$$\boldsymbol{\tau} = \boldsymbol{\tau}_L + \boldsymbol{\tau}_R + \boldsymbol{\tau}_T + \boldsymbol{\tau}_B$$



Torque on a Current Carrying Loop in a Magnetic Field

The torque on the left side is

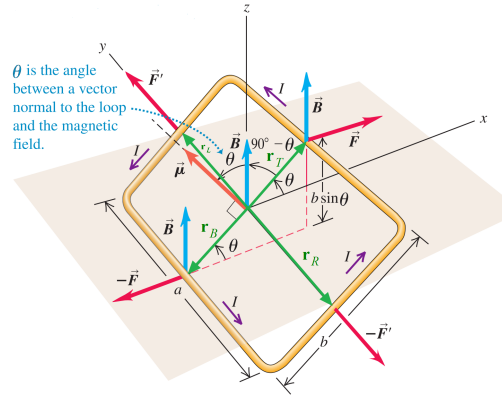
$$\boldsymbol{\tau}_L = \mathbf{r}_L \times \mathbf{F}_L$$



Torque on a Current Carrying Loop in a Magnetic Field

The torque on the left side is

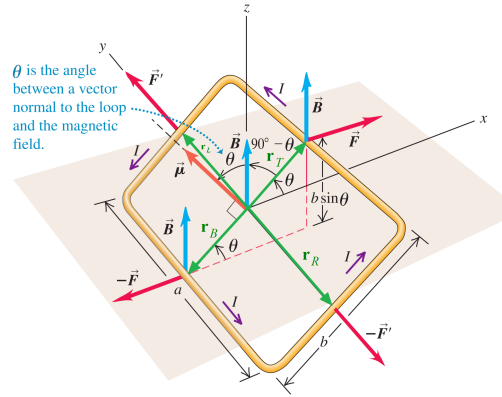
$$\boldsymbol{\tau}_L = \mathbf{r}_L \times \mathbf{F}_L = \left[\frac{1}{2} \mathbf{L}_T \right] \times [I \mathbf{L}_L \times \mathbf{B}]$$



Torque on a Current Carrying Loop in a Magnetic Field

The torque on the left side is

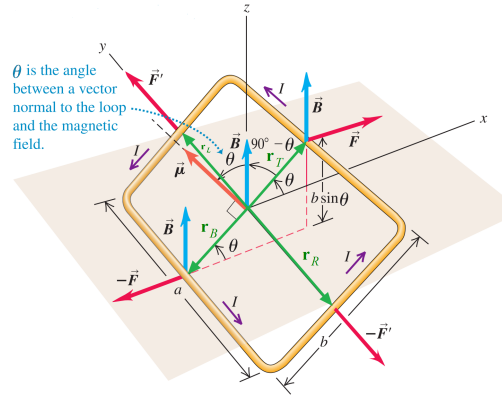
$$\begin{aligned}\boldsymbol{\tau}_L &= \mathbf{r}_L \times \mathbf{F}_L = \left[\frac{1}{2} \mathbf{L}_T \right] \times [I \mathbf{L}_L \times \mathbf{B}] \\ &= \frac{1}{2} I \underbrace{\mathbf{L}_T \times \mathbf{L}_L}_0 \times \mathbf{B}\end{aligned}$$



Torque on a Current Carrying Loop in a Magnetic Field

The torque on the left side is

$$\begin{aligned}\tau_L &= \mathbf{r}_L \times \mathbf{F}_L = \left[\frac{1}{2} \mathbf{L}_T \right] \times [I \mathbf{L}_L \times \mathbf{B}] \\ &= \frac{1}{2} I \underbrace{\mathbf{L}_T \times \mathbf{L}_L}_0 \times \mathbf{B} \\ &= 0\end{aligned}$$



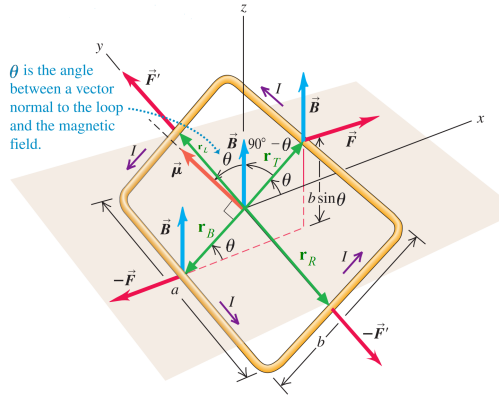
Torque on a Current Carrying Loop in a Magnetic Field

The torque on the left side is

$$\begin{aligned}\tau_L &= \mathbf{r}_L \times \mathbf{F}_L = \left[\frac{1}{2} \mathbf{L}_T \right] \times [I \mathbf{L}_L \times \mathbf{B}] \\ &= \frac{1}{2} I \underbrace{\mathbf{L}_T \times \mathbf{L}_L}_0 \times \mathbf{B} \\ &= 0\end{aligned}$$

The torque on the left side is (use $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$)

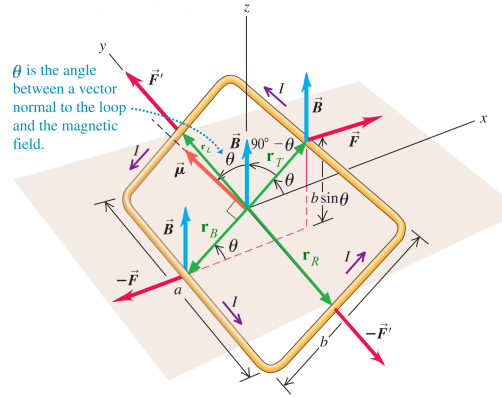
$$\begin{aligned}\tau_L &= \mathbf{r}_L \times \mathbf{F}_L = \left[\frac{1}{2} \mathbf{L}_T \right] \times [I \mathbf{L}_L \times \mathbf{B}] \\ &= \frac{1}{2} I \left[\mathbf{L}_L \underbrace{(\mathbf{L}_T \cdot \mathbf{B})}_0 - \mathbf{B} \underbrace{(\mathbf{L}_T \cdot \mathbf{L}_L)}_0 \right] \\ &= 0\end{aligned}$$



Torque on a Current Carrying Loop in a Magnetic Field

The torque on the top side is

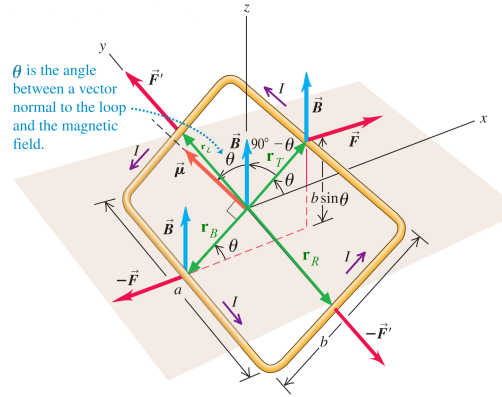
$$\boldsymbol{\tau}_T = \mathbf{r}_T \times \mathbf{F}_T$$



Torque on a Current Carrying Loop in a Magnetic Field

The torque on the top side is

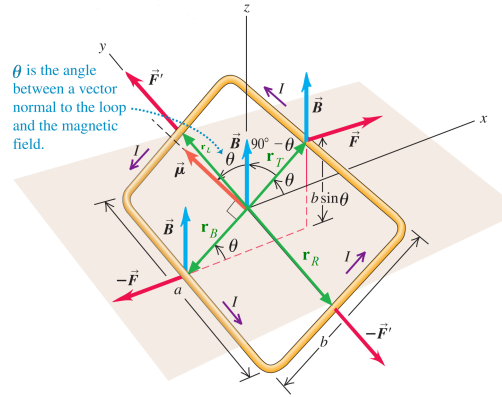
$$\boldsymbol{\tau}_T = \mathbf{r}_T \times \mathbf{F}_T = \mathbf{r}_T \times (I \mathbf{L}_T \times \mathbf{B})$$



Torque on a Current Carrying Loop in a Magnetic Field

The torque on the top side is

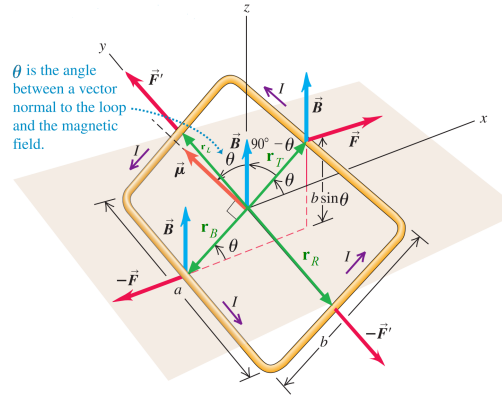
$$\begin{aligned}\boldsymbol{\tau}_T &= \mathbf{r}_T \times \mathbf{F}_T = \mathbf{r}_T \times (I\mathbf{L}_T \times \mathbf{B}) \\ &= \left(\frac{1}{2} \mathbf{L}_R \right) \times (I\mathbf{L}_T \times \mathbf{B})\end{aligned}$$



Torque on a Current Carrying Loop in a Magnetic Field

The torque on the top side is

$$\begin{aligned}\boldsymbol{\tau}_T &= \mathbf{r}_T \times \mathbf{F}_T = \mathbf{r}_T \times (I \mathbf{L}_T \times \mathbf{B}) \\ &= \left(\frac{1}{2} \mathbf{L}_R \right) \times (I \mathbf{L}_T \times \mathbf{B}) \\ &= \frac{1}{2} I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B}\end{aligned}$$



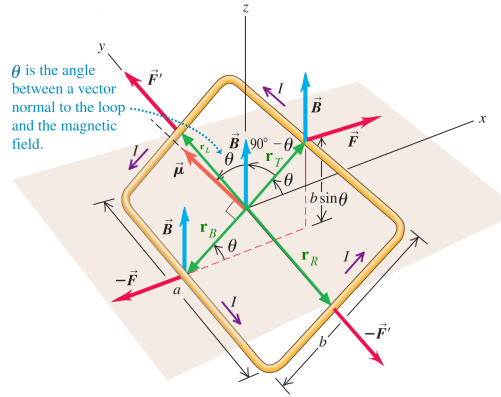
Torque on a Current Carrying Loop in a Magnetic Field

The torque on the top side is

$$\begin{aligned}\tau_T &= \mathbf{r}_T \times \mathbf{F}_T = \mathbf{r}_T \times (I \mathbf{L}_T \times \mathbf{B}) \\ &= \left(\frac{1}{2} \mathbf{L}_R \right) \times (I \mathbf{L}_T \times \mathbf{B}) \\ &= \frac{1}{2} I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B}\end{aligned}$$

The torque on the bottom side is

$$\tau_B = \mathbf{r}_B \times \mathbf{F}_B$$



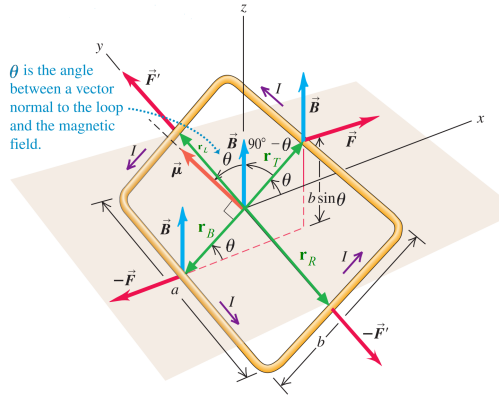
Torque on a Current Carrying Loop in a Magnetic Field

The torque on the top side is

$$\begin{aligned}\tau_T &= \mathbf{r}_T \times \mathbf{F}_T = \mathbf{r}_T \times (I \mathbf{L}_T \times \mathbf{B}) \\ &= \left(\frac{1}{2} \mathbf{L}_R \right) \times (I \mathbf{L}_T \times \mathbf{B}) \\ &= \frac{1}{2} I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B}\end{aligned}$$

The torque on the bottom side is

$$\tau_B = \mathbf{r}_B \times \mathbf{F}_B = (-\mathbf{r}_T) \times (-\mathbf{F}_T)$$



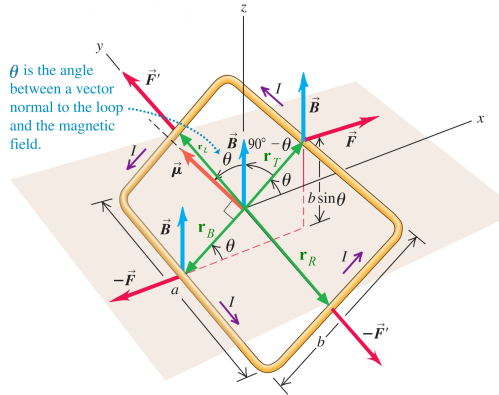
Torque on a Current Carrying Loop in a Magnetic Field

The torque on the top side is

$$\begin{aligned}\tau_T &= \mathbf{r}_T \times \mathbf{F}_T = \mathbf{r}_T \times (I\mathbf{L}_T \times \mathbf{B}) \\ &= \left(\frac{1}{2} \mathbf{L}_R \right) \times (I\mathbf{L}_T \times \mathbf{B}) \\ &= \frac{1}{2} I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B}\end{aligned}$$

The torque on the bottom side is

$$\begin{aligned}\tau_B &= \mathbf{r}_B \times \mathbf{F}_B = (-\mathbf{r}_T) \times (-\mathbf{F}_T) \\ &= \mathbf{r}_T \times \mathbf{F}_T\end{aligned}$$



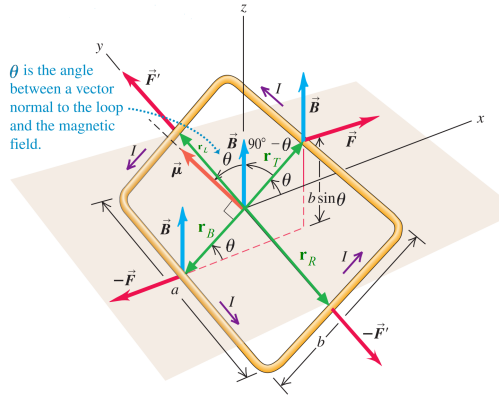
Torque on a Current Carrying Loop in a Magnetic Field

The torque on the top side is

$$\begin{aligned}\boldsymbol{\tau}_T &= \mathbf{r}_T \times \mathbf{F}_T = \mathbf{r}_T \times (I\mathbf{L}_T \times \mathbf{B}) \\ &= \left(\frac{1}{2} \mathbf{L}_R \right) \times (I\mathbf{L}_T \times \mathbf{B}) \\ &= \frac{1}{2} I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B}\end{aligned}$$

The torque on the bottom side is

$$\begin{aligned}\boldsymbol{\tau}_B &= \mathbf{r}_B \times \mathbf{F}_B = (-\mathbf{r}_T) \times (-\mathbf{F}_T) \\ &= \mathbf{r}_T \times \mathbf{F}_T \\ &= \boldsymbol{\tau}_T\end{aligned}$$



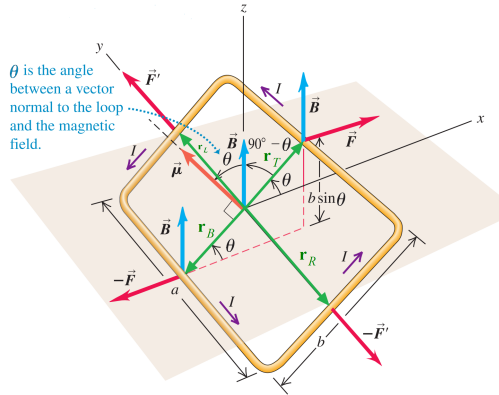
Torque on a Current Carrying Loop in a Magnetic Field

The torque on the top side is

$$\begin{aligned}\tau_T &= \mathbf{r}_T \times \mathbf{F}_T = \mathbf{r}_T \times (I \mathbf{L}_T \times \mathbf{B}) \\ &= \left(\frac{1}{2} \mathbf{L}_R \right) \times (I \mathbf{L}_T \times \mathbf{B}) \\ &= \frac{1}{2} I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B}\end{aligned}$$

The torque on the bottom side is

$$\begin{aligned}\tau_B &= \mathbf{r}_B \times \mathbf{F}_B = (-\mathbf{r}_T) \times (-\mathbf{F}_T) \\ &= \mathbf{r}_T \times \mathbf{F}_T \\ &= \tau_T \\ &= \frac{1}{2} I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B}\end{aligned}$$



Torque on a Current Carrying Loop in a Magnetic Field

So the total torque is

$$\tau = \tau_L + \tau_R + \tau_T + \tau_B$$

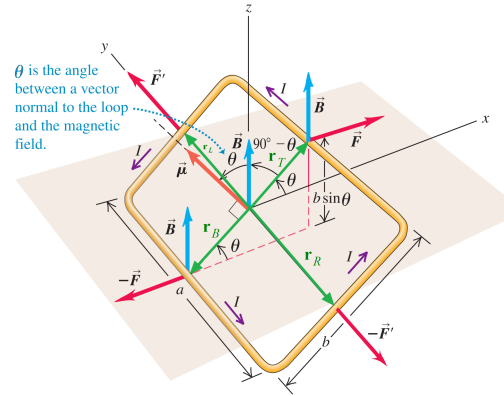
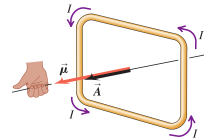


Figure 27.32 The right-hand rule determines the direction of the magnetic moment of a current-carrying loop. This is also the direction of the loop's area vector $\vec{A}$; $\vec{\mu} = I\vec{A}$ is a vector equation.



Torque on a Current Carrying Loop in a Magnetic Field

So the total torque is

$$\tau = \tau_L + \tau_R + \tau_T + \tau_B = 2\tau_T$$

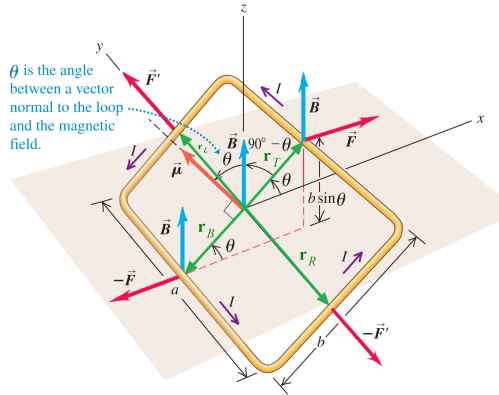
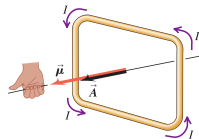


Figure 27.32 The right-hand rule determines the direction of the magnetic moment of a current-carrying loop. This is also the direction of the loop's area vector $\vec{A}$; $\vec{\mu} = I\vec{A}$ is a vector equation.



Torque on a Current Carrying Loop in a Magnetic Field

So the total torque is

$$\begin{aligned}\tau &= \tau_L + \tau_R + \tau_T + \tau_B = 2\tau_T \\ &= 2 \left(\frac{1}{2} I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B} \right)\end{aligned}$$

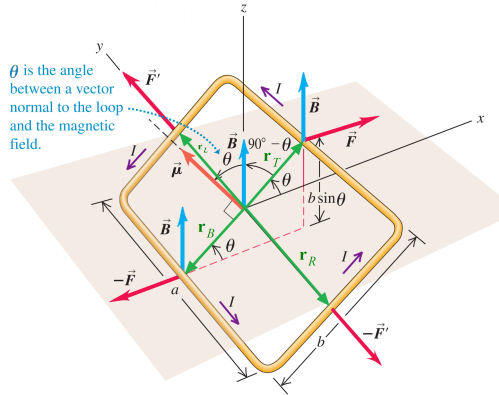
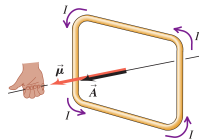


Figure 27.32 The right-hand rule determines the direction of the magnetic moment of a current-carrying loop. This is also the direction of the loop's area vector $\vec{A}$; $\vec{\mu} = I\vec{A}$ is a vector equation.



Torque on a Current Carrying Loop in a Magnetic Field

So the total torque is

$$\begin{aligned}\tau &= \tau_L + \tau_R + \tau_T + \tau_B = 2\tau_T \\ &= 2 \left(\frac{1}{2} I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B} \right) \\ &= I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B}\end{aligned}$$

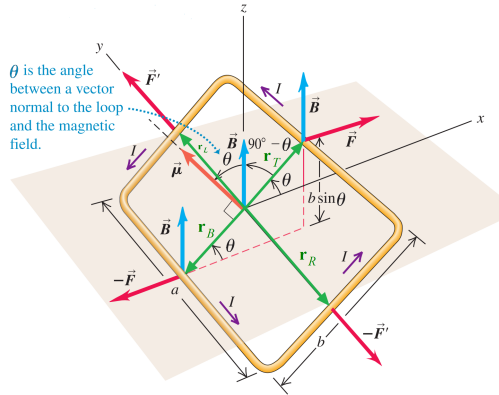
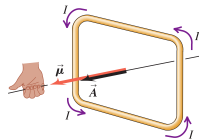


Figure 27.32 The right-hand rule determines the direction of the magnetic moment of a current-carrying loop. This is also the direction of the loop's area vector $\vec{A}$; $\vec{\mu} = I\vec{A}$ is a vector equation.



Torque on a Current Carrying Loop in a Magnetic Field

So the total torque is

$$\begin{aligned}\tau &= \tau_L + \tau_R + \tau_T + \tau_B = 2\tau_T \\ &= 2 \left(\frac{1}{2} I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B} \right) \\ &= I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B}\end{aligned}$$

But

$$\mathbf{A} = \mathbf{L}_R \times \mathbf{L}_T$$

is the area vector of the loop magnitude $L_R L_T = L_L L_B$ and its direction is along the thumb if four fingers rotate along the current I .

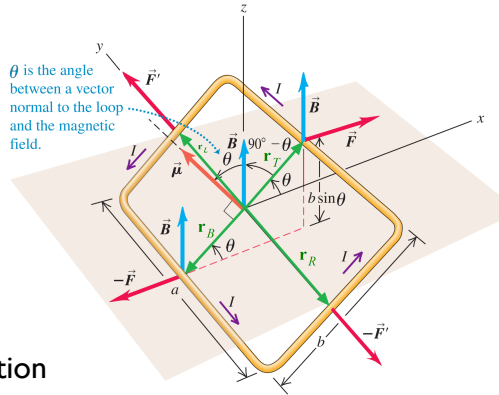
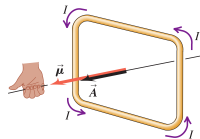


Figure 27.32 The right-hand rule determines the direction of the magnetic moment of a current-carrying loop. This is also the direction of the loop's area vector $\vec{A}$; $\vec{\mu} = I\vec{A}$ is a vector equation.



Torque on a Current Carrying Loop in a Magnetic Field

So the total torque is

$$\begin{aligned}\tau &= \tau_L + \tau_R + \tau_T + \tau_B = 2\tau_T \\ &= 2 \left(\frac{1}{2} I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B} \right) \\ &= I \mathbf{L}_R \times \mathbf{L}_T \times \mathbf{B}\end{aligned}$$

But

$$\mathbf{A} = \mathbf{L}_R \times \mathbf{L}_T$$

is the area vector of the loop magnitude $L_R L_T = L_L L_B$ and its direction is along the thumb if four fingers rotate along the current I .

This means the total torque on the loop can be written as

$$\tau_T = I \mathbf{A} \times \mathbf{B}$$

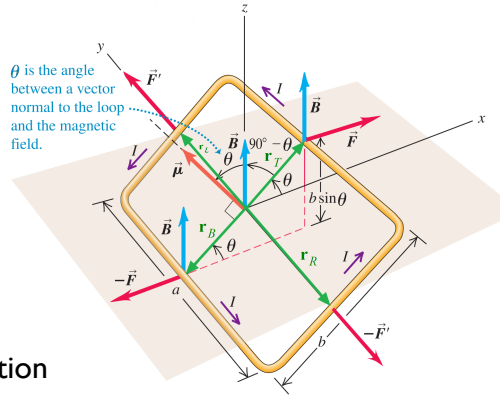
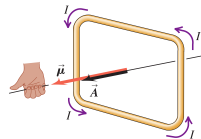


Figure 27.32 The right-hand rule determines the direction of the magnetic moment of a current-carrying loop. This is also the direction of the loop's area vector $\vec{A}$; $\vec{\mu} = I\vec{A}$ is a vector equation.



Torque on a Current Carrying Loop in a Magnetic Field

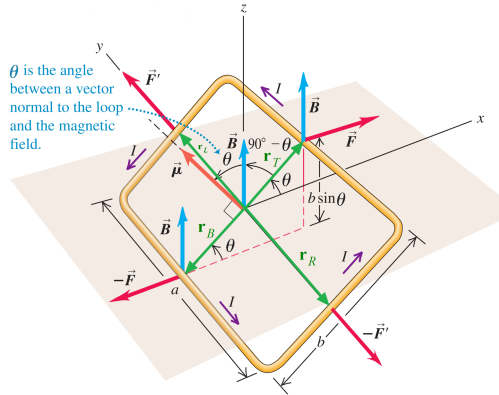
We define the **magnetic dipole moment** or **magnetic moment** of the loop as

$$\vec{\mu} = I\vec{A}$$

where

$$\vec{A} = \vec{L}_R \times \vec{L}_T$$

is the area vector of the loop



Torque on a Current Carrying Loop in a Magnetic Field

We define the **magnetic dipole moment** or **magnetic moment** of the loop as

$$\mu = I\mathbf{A}$$

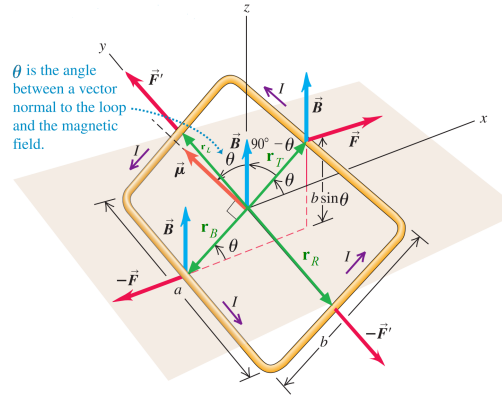
where

$$\mathbf{A} = \mathbf{L}_R \times \mathbf{L}_T$$

is the area vector of the loop

Now the total torque on the loop can be written as

$$\tau = \mu \times \mathbf{B}$$



Torque on a Current Carrying Loop in a Magnetic Field

We define the **magnetic dipole moment** or **magnetic moment** of the loop as

$$\mu = I\mathbf{A}$$

where

$$\mathbf{A} = \mathbf{L}_R \times \mathbf{L}_T$$

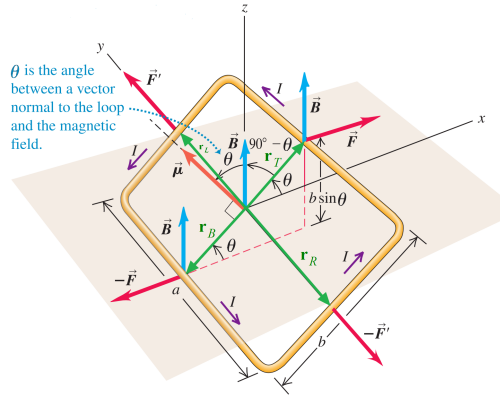
is the area vector of the loop

Now the total torque on the loop can be written as

$$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$$

Its magnitude is

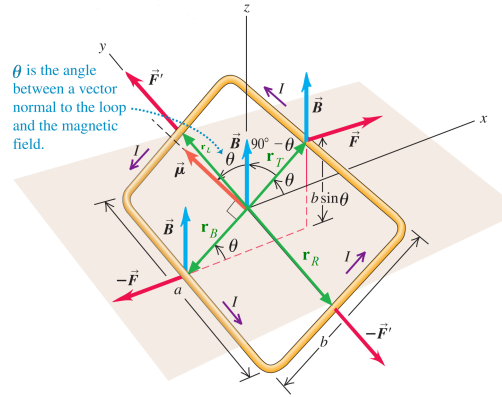
$$\tau = \mu B \sin(\theta_{\mu, \mathbf{B}}) = IAB \sin(\theta_{\mu, \mathbf{B}})$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

The potential energy of each side of the loop upon a rotation (against the angle θ seen in the picture) with displacement $d\mathbf{r}$ is

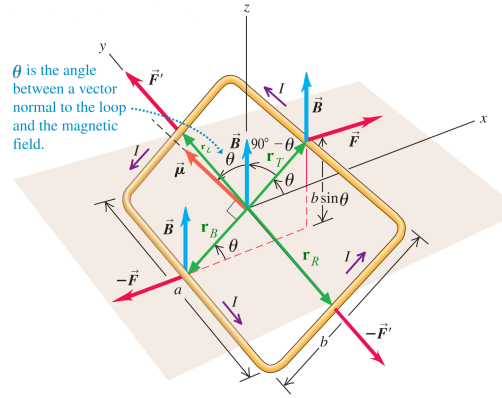
$$dU = -dW$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

The potential energy of each side of the loop upon a rotation (against the angle θ seen in the picture) with displacement $d\mathbf{r}$ is

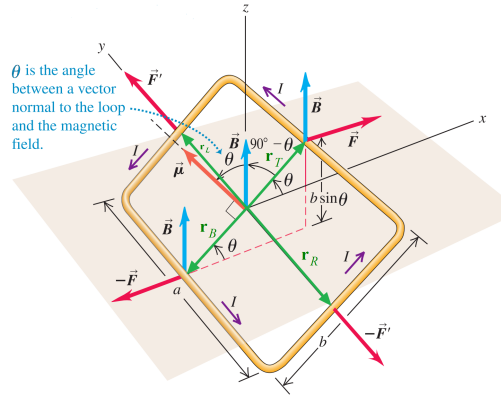
$$dU = -dW = -\mathbf{F} \cdot d\mathbf{r}$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

The potential energy of each side of the loop upon a rotation (against the angle θ seen in the picture) with displacement $d\mathbf{r}$ is

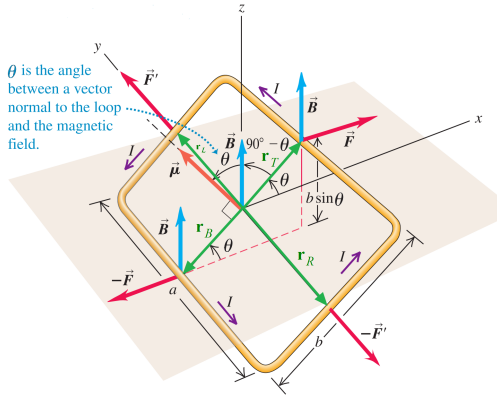
$$dU = -dW = -\mathbf{F} \cdot d\mathbf{r} = -I(\mathbf{L} \times \mathbf{B}) \cdot d\mathbf{r}$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

The potential energy of each side of the loop upon a rotation (against the angle θ seen in the picture) with displacement $d\mathbf{r}$ is

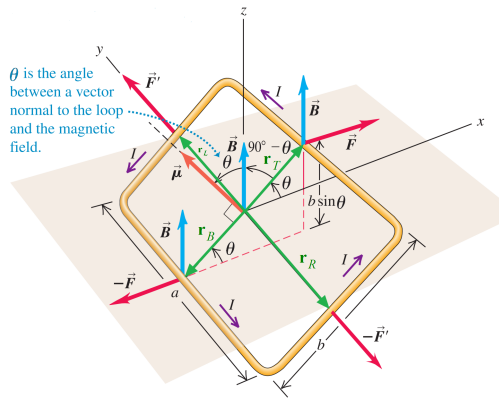
$$\begin{aligned}dU &= -dW = -\mathbf{F} \cdot d\mathbf{r} = -I(\mathbf{L} \times \mathbf{B}) \cdot d\mathbf{r} \\ &= -I d\mathbf{r} \cdot (\mathbf{L} \times \mathbf{B})\end{aligned}$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

The potential energy of each side of the loop upon a rotation (against the angle θ seen in the picture) with displacement $d\mathbf{r}$ is

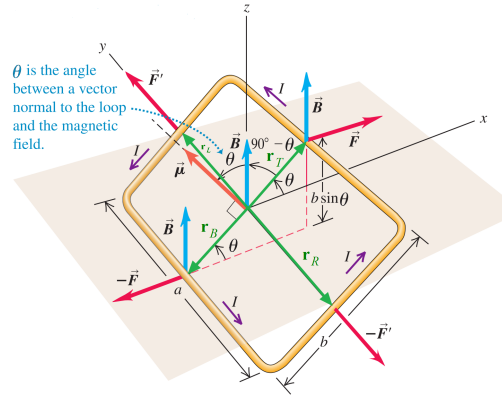
$$\begin{aligned}dU &= -dW = -\mathbf{F} \cdot d\mathbf{r} = -I(\mathbf{L} \times \mathbf{B}) \cdot d\mathbf{r} \\&= -I d\mathbf{r} \cdot (\mathbf{L} \times \mathbf{B}) \\&= -I(d\mathbf{r} \times \mathbf{L}) \cdot \mathbf{B}\end{aligned}$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

So for each side we have

$$dU_L = -I \left(\underbrace{d\mathbf{r}_L}_0 \times \mathbf{L}_L \right) \cdot \mathbf{B} = 0$$

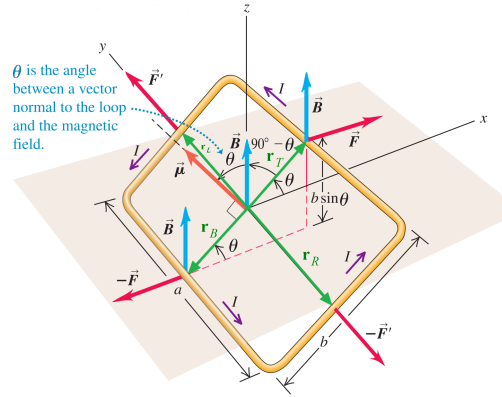


Potential Energy of a Current Carrying Loop in a Magnetic Field

So for each side we have

$$dU_L = -I \left(\underbrace{d\mathbf{r}_L}_0 \times \mathbf{L}_L \right) \cdot \mathbf{B} = 0$$

$$dU_R = -I \left(\underbrace{d\mathbf{r}_R}_0 \times \mathbf{L}_R \right) \cdot \mathbf{B} = 0$$



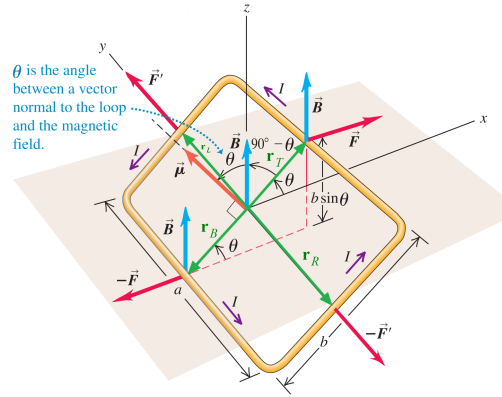
Potential Energy of a Current Carrying Loop in a Magnetic Field

So for each side we have

$$dU_L = -I \left(\underbrace{d\mathbf{r}_L}_0 \times \mathbf{L}_L \right) \cdot \mathbf{B} = 0$$

$$dU_R = -I \left(\underbrace{d\mathbf{r}_R}_0 \times \mathbf{L}_R \right) \cdot \mathbf{B} = 0$$

$$dU_T = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B}$$



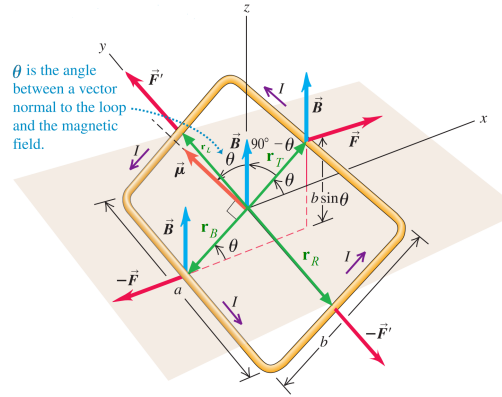
Potential Energy of a Current Carrying Loop in a Magnetic Field

So for each side we have

$$dU_L = -I \left(\underbrace{d\mathbf{r}_L}_0 \times \mathbf{L}_L \right) \cdot \mathbf{B} = 0$$

$$dU_R = -I \left(\underbrace{d\mathbf{r}_R}_0 \times \mathbf{L}_R \right) \cdot \mathbf{B} = 0$$

$$dU_T = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B} = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B}$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

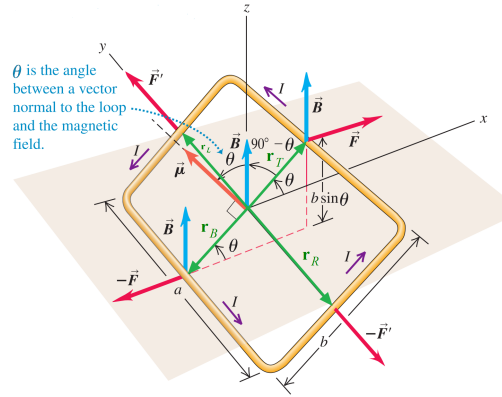
So for each side we have

$$dU_L = -I \left(\underbrace{d\mathbf{r}_L}_0 \times \mathbf{L}_L \right) \cdot \mathbf{B} = 0$$

$$dU_R = -I \left(\underbrace{d\mathbf{r}_R}_0 \times \mathbf{L}_R \right) \cdot \mathbf{B} = 0$$

$$dU_T = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B} = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B}$$

$$dU_B = -I (d\mathbf{r}_B \times \mathbf{L}_B) \cdot \mathbf{B}$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

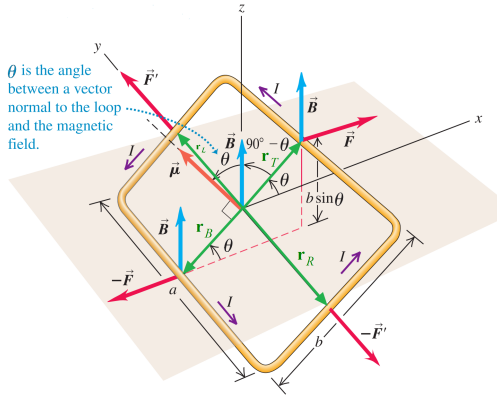
So for each side we have

$$dU_L = -I \left(\underbrace{d\mathbf{r}_L}_0 \times \mathbf{L}_L \right) \cdot \mathbf{B} = 0$$

$$dU_R = -I \left(\underbrace{d\mathbf{r}_R}_0 \times \mathbf{L}_R \right) \cdot \mathbf{B} = 0$$

$$dU_T = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B} = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B}$$

$$dU_B = -I (d\mathbf{r}_B \times \mathbf{L}_B) \cdot \mathbf{B} = -I [(-d\mathbf{r}_T) \times (-\mathbf{L}_T)] \cdot \mathbf{B}$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

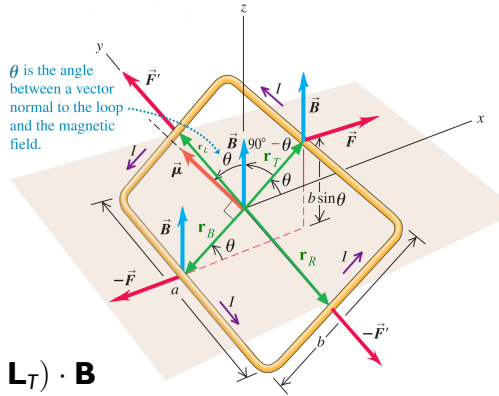
So for each side we have

$$dU_L = -I \left(\underbrace{d\mathbf{r}_L}_0 \times \mathbf{L}_L \right) \cdot \mathbf{B} = 0$$

$$dU_R = -I \left(\underbrace{d\mathbf{r}_R}_0 \times \mathbf{L}_R \right) \cdot \mathbf{B} = 0$$

$$dU_T = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B} = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B}$$

$$dU_B = -I (d\mathbf{r}_B \times \mathbf{L}_B) \cdot \mathbf{B} = -I [(-d\mathbf{r}_T) \times (-\mathbf{L}_T)] \cdot \mathbf{B} = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B}$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

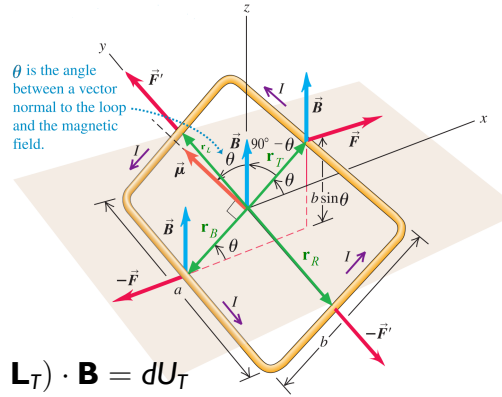
So for each side we have

$$dU_L = -I \left(\underbrace{d\mathbf{r}_L}_0 \times \mathbf{L}_L \right) \cdot \mathbf{B} = 0$$

$$dU_R = -I \left(\underbrace{d\mathbf{r}_R}_0 \times \mathbf{L}_R \right) \cdot \mathbf{B} = 0$$

$$dU_T = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B} = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B}$$

$$dU_B = -I (d\mathbf{r}_B \times \mathbf{L}_B) \cdot \mathbf{B} = -I [(-d\mathbf{r}_T) \times (-\mathbf{L}_T)] \cdot \mathbf{B} = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B} = dU_T$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

So for each side we have

$$dU_L = -I \left(\underbrace{d\mathbf{r}_L}_0 \times \mathbf{L}_L \right) \cdot \mathbf{B} = 0$$

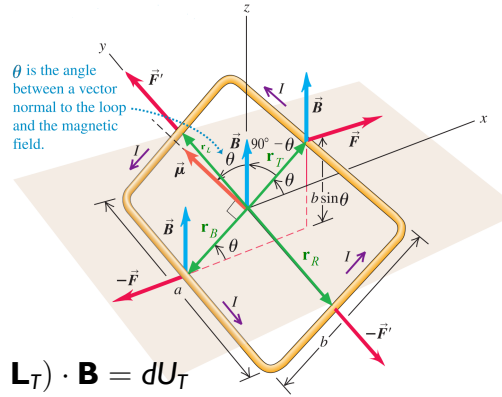
$$dU_R = -I \left(\underbrace{d\mathbf{r}_R}_0 \times \mathbf{L}_R \right) \cdot \mathbf{B} = 0$$

$$dU_T = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B} = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B}$$

$$dU_B = -I (d\mathbf{r}_B \times \mathbf{L}_B) \cdot \mathbf{B} = -I [(-d\mathbf{r}_T) \times (-\mathbf{L}_T)] \cdot \mathbf{B} = -I (d\mathbf{r}_T \times \mathbf{L}_T) \cdot \mathbf{B} = dU_T$$

So the total infinitesimal change in potential energy is

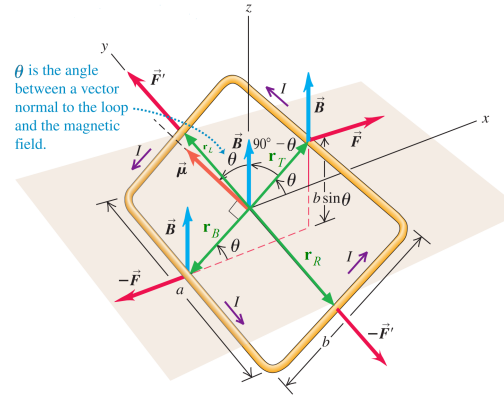
$$dU = dU_L + dU_R + dU_T + dU_B = 2dU_T$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

Now we find the total U keeping in mind that I , $\mathbf{L}$ and $\mathbf{B}$ are constants

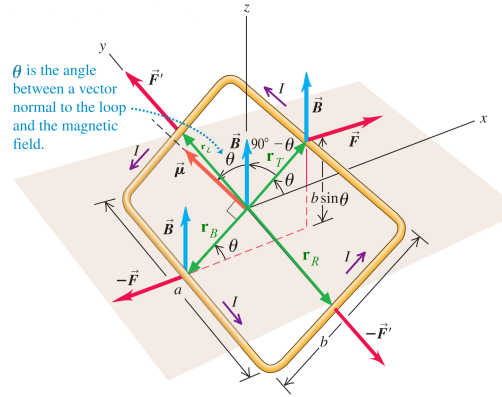
$$U = \int dU$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

Now we find the total U keeping in mind that I , $\mathbf{L}$ and $\mathbf{B}$ are constants

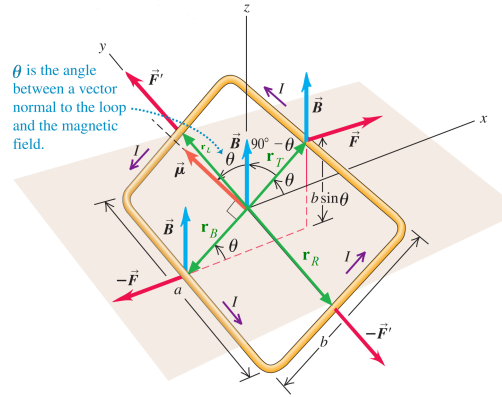
$$U = \int dU = 2 \int dU_T$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

Now we find the total U keeping in mind that I , $\mathbf{L}$ and $\mathbf{B}$ are constants

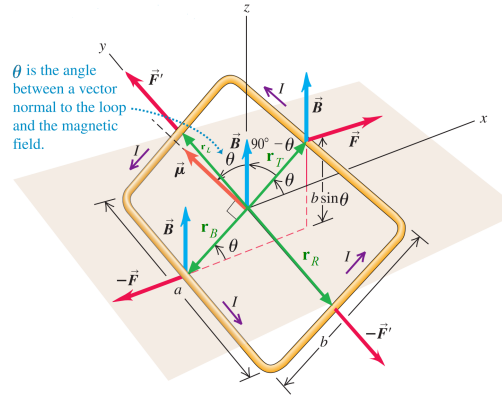
$$U = \int dU = 2 \int dU_T = -2I \left[\left(\int d\mathbf{r}_T \right) \times \mathbf{L}_T \right] \cdot \mathbf{B}$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

Now we find the total U keeping in mind that I , $\mathbf{L}$ and $\mathbf{B}$ are constants

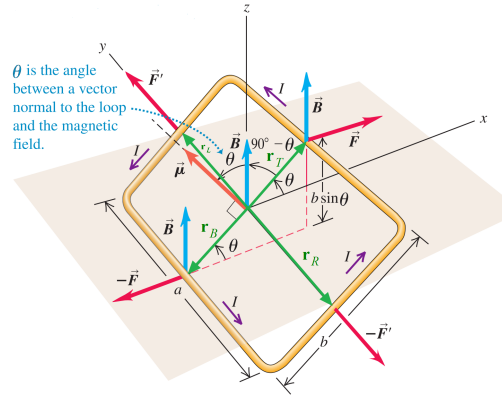
$$\begin{aligned} U &= \int dU = 2 \int dU_T = -2I \left[\left(\int d\mathbf{r}_T \right) \times \mathbf{L}_T \right] \cdot \mathbf{B} \\ &= -2I [\mathbf{r}_T \times \mathbf{L}_T] \cdot \mathbf{B} \end{aligned}$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

Now we find the total U keeping in mind that I , $\mathbf{L}$ and $\mathbf{B}$ are constants

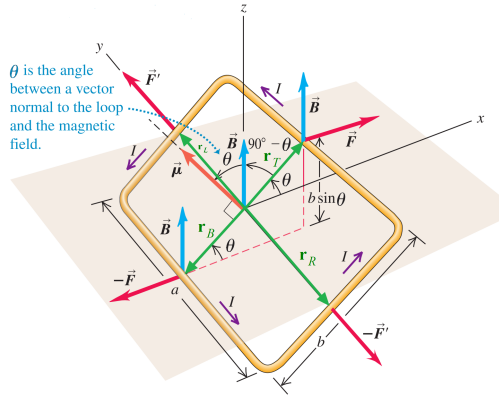
$$\begin{aligned} U &= \int dU = 2 \int dU_T = -2I \left[\left(\int d\mathbf{r}_T \right) \times \mathbf{L}_T \right] \cdot \mathbf{B} \\ &= -2I [\mathbf{r}_T \times \mathbf{L}_T] \cdot \mathbf{B} \\ &= -2I \left[\frac{1}{2} \underbrace{\mathbf{L}_R \times \mathbf{L}_T}_{\mathbf{A}} \right] \cdot \mathbf{B} \end{aligned}$$



Potential Energy of a Current Carrying Loop in a Magnetic Field

Now we find the total U keeping in mind that I , $\mathbf{L}$ and $\mathbf{B}$ are constants

$$\begin{aligned} U &= \int dU = 2 \int dU_T = -2I \left[\left(\int d\mathbf{r}_T \right) \times \mathbf{L}_T \right] \cdot \mathbf{B} \\ &= -2I [\mathbf{r}_T \times \mathbf{L}_T] \cdot \mathbf{B} \\ &= -2I \left[\frac{1}{2} \underbrace{\mathbf{L}_R \times \mathbf{L}_T}_{\mathbf{A}} \right] \cdot \mathbf{B} \\ &= - \underbrace{I\mathbf{A}}_{\mu} \cdot \mathbf{B} \end{aligned}$$



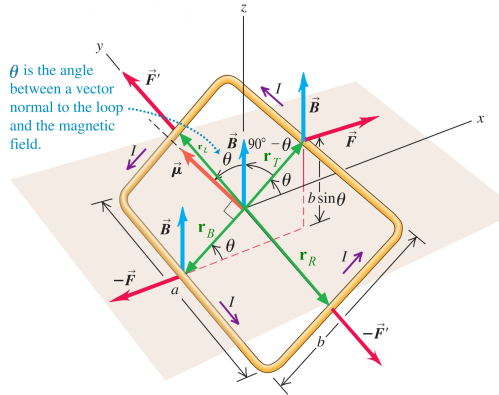
Potential Energy of a Current Carrying Loop in a Magnetic Field

Now we find the total U keeping in mind that I , $\mathbf{L}$ and $\mathbf{B}$ are constants

$$\begin{aligned} U &= \int dU = 2 \int dU_T = -2I \left[\left(\int d\mathbf{r}_T \right) \times \mathbf{L}_T \right] \cdot \mathbf{B} \\ &= -2I [\mathbf{r}_T \times \mathbf{L}_T] \cdot \mathbf{B} \\ &= -2I \left[\frac{1}{2} \underbrace{\mathbf{L}_R \times \mathbf{L}_T}_{\mathbf{A}} \right] \cdot \mathbf{B} \\ &= - \underbrace{I\mathbf{A}}_{\boldsymbol{\mu}} \cdot \mathbf{B} \end{aligned}$$

Thus we conclude that the potential energy of loop with current I in a magnetic field $\mathbf{B}$ is

$$U = -\boldsymbol{\mu} \cdot \mathbf{B}$$

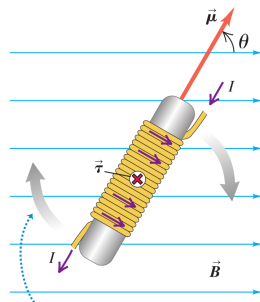


Magnetic Force on a Coil

If we have a coil with N loops (e.g., a solenoid), the total torque on it in a magnetic field $\mathbf{B}$ is N times the torque on one loop

$$\tau_N = N\tau = N\mu \times \mathbf{B}$$

Figure 27.34 The torque $\vec{\tau} = \vec{\mu} \times \vec{B}$ on this solenoid in a uniform magnetic field is directed straight into the page. An actual solenoid has many more turns, wrapped closely together.



The torque tends to make the solenoid rotate clockwise in the plane of the page, aligning magnetic moment $\vec{\mu}$ with field $\vec{B}$.

Magnetic Force on a Coil

If we have a coil with N loops (e.g., a solenoid), the total torque on it in a magnetic field $\mathbf{B}$ is N times the torque on one loop

$$\tau_N = N\tau = N\mu \times \mathbf{B}$$

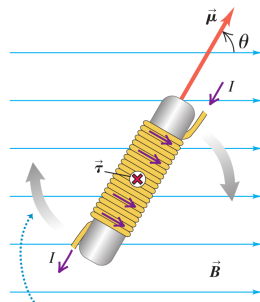
We can define the μ_N of the loop as

$$\mu_N = N\mu$$

and the total torque would be

$$\tau_N = \mu_N \times \mathbf{B}$$

Figure 27.34 The torque $\vec{\tau} = \vec{\mu} \times \vec{B}$ on this solenoid in a uniform magnetic field is directed straight into the page. An actual solenoid has many more turns, wrapped closely together.

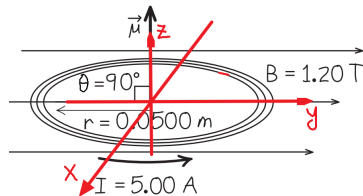


The torque tends to make the solenoid rotate clockwise in the plane of the page, aligning magnetic moment $\vec{\mu}$ with field $\vec{B}$.

Exercise 1

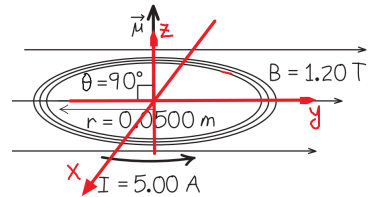
<https://epoll.srv.ualberta.ca> – Code: MBD

A circular coil 0.05 m in radius, with 30 turns of wire, lies in a horizontal plane. It carries a counterclockwise (as viewed from above) current of 5 A. The coil is in a uniform 1.2 T magnetic field directed toward the right. (a) Find the magnetic moment vector and the torque vector on the coil. (b) If the coil rotates from its initial orientation to one in which its magnetic moment μ is parallel to $\mathbf{B}$, what is the change in potential energy?



Exercise I

<https://epoll.srv.ualberta.ca> – Code: MBD



Outline

7 Magnetic Field and Magnetic Forces

- Magnetic Force and Torque on a Current-carrying Conductor (Continued)

8 Sources of Magnetic Field

- Magnetic Field of a Moving Charge

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

Outline

7 Magnetic Field and Magnetic Forces

- Magnetic Force and Torque on a Current-carrying Conductor (Continued)

8 Sources of Magnetic Field

- Magnetic Field of a Moving Charge

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

Vector Magnetic Field of a Moving Charge

The magnetic field $\mathbf{B}$ due to a point charge q moving at a constant velocity $\mathbf{v}$ at a point P a distance vector $\mathbf{r}$ from q is

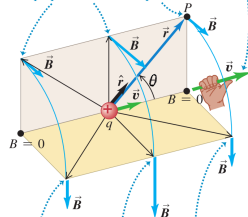
$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \mathbf{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \hat{\mathbf{r}}}{r^2}$$

where $\mathbf{r}$ points from source q to the point P

(a) Perspective view

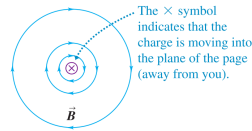
Right-hand rule for the magnetic field due to a positive charge moving at constant velocity: Point the thumb of your right hand in the direction of the velocity. Your fingers now curl around the charge in the direction of the magnetic field lines. (If the charge is negative, the field lines are in the opposite direction.)

For these field points, $\vec{r}$ and $\vec{v}$ both lie in the beige plane, and $\vec{B}$ is perpendicular to this plane.



For these field points, $\vec{r}$ and $\vec{v}$ both lie in the gold plane, and $\vec{B}$ is perpendicular to this plane.

(b) View from behind the charge



Vector Magnetic Field of a Moving Charge

The magnetic field $\mathbf{B}$ due to a point charge q moving at a constant velocity $\mathbf{v}$ at a point P a distance vector $\mathbf{r}$ from q is

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \mathbf{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \hat{\mathbf{r}}}{r^2}$$

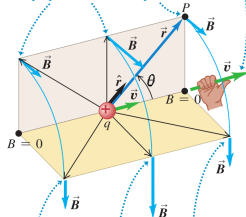
where $\mathbf{r}$ points from source q to the point P

The universal constant $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$ is called the **magnetic constant** (has nothing to do with magnetic moment μ)

(a) Perspective view

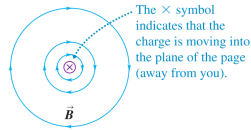
Right-hand rule for the magnetic field due to a positive charge moving at constant velocity: Point the thumb of your right hand in the direction of the velocity. Your fingers now curl around the charge in the direction of the magnetic field lines. (If the charge is negative, the field lines are in the opposite direction.)

For these field points, $\vec{r}$ and $\vec{v}$ both lie in the beige plane, and $\vec{B}$ is perpendicular to this plane.



For these field points, $\vec{r}$ and $\vec{v}$ both lie in the gold plane, and $\vec{B}$ is perpendicular to this plane.

(b) View from behind the charge



Vector Magnetic Field of a Moving Charge

The magnetic field $\mathbf{B}$ due to a point charge q moving at a constant velocity $\mathbf{v}$ at a point P a distance vector $\mathbf{r}$ from q is

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \mathbf{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \hat{\mathbf{r}}}{r^2}$$

where $\mathbf{r}$ points from source q to the point P

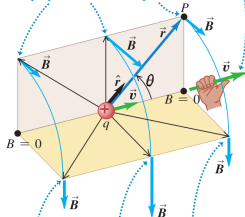
The universal constant $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$ is called the **magnetic constant** (has nothing to do with magnetic moment μ)

For a point charge moving with velocity $\mathbf{v}$, the magnetic field lines are circles centered on the line of $\mathbf{v}$ and lying in planes perpendicular to this line

(a) Perspective view

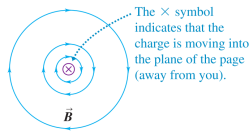
Right-hand rule for the magnetic field due to a positive charge moving at constant velocity: Point the thumb of your right hand in the direction of the velocity. Your fingers now curl around the charge in the direction of the magnetic field lines. (If the charge is negative, the field lines are in the opposite direction.)

For these field points, $\vec{r}$ and $\vec{v}$ both lie in the beige plane, and $\vec{B}$ is perpendicular to this plane.



For these field points, $\vec{r}$ and $\vec{v}$ both lie in the gold plane, and $\vec{B}$ is perpendicular to this plane.

(b) View from behind the charge



Vector Magnetic Field of a Moving Charge

The magnetic field $\mathbf{B}$ due to a point charge q moving at a constant velocity $\mathbf{v}$ at a point P a distance vector $\mathbf{r}$ from q is

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \mathbf{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \hat{\mathbf{r}}}{r^2}$$

where $\mathbf{r}$ points from source q to the point P

The universal constant $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$ is called the **magnetic constant** (has nothing to do with magnetic moment μ)

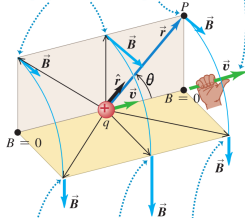
For a point charge moving with velocity $\mathbf{v}$, the magnetic field lines are circles centered on the line of $\mathbf{v}$ and lying in planes perpendicular to this line

For a positive point charge moving with velocity $\mathbf{v}$, the magnetic field lines are found by the right hand rule: thumb in the direction of $\mathbf{v}$ and direction of rotation of other four fingers are the direction of the field lines

(a) Perspective view

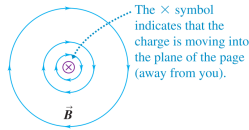
Right-hand rule for the magnetic field due to a positive charge moving at constant velocity: Point the thumb of your right hand in the direction of the velocity. Your fingers now curl around the charge in the direction of the magnetic field lines. (If the charge is negative, the field lines are in the opposite direction.)

For these field points, $\vec{r}$ and $\vec{v}$ both lie in the beige plane, and $\vec{B}$ is perpendicular to this plane.



For these field points, $\vec{r}$ and $\vec{v}$ both lie in the gold plane, and $\vec{B}$ is perpendicular to this plane.

(b) View from behind the charge



Vector Magnetic Field of a Moving Charge

The magnetic field $\mathbf{B}$ due to a point charge q moving at a constant velocity $\mathbf{v}$ at a point P a distance vector $\mathbf{r}$ from q is

$$\mathbf{B} = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \mathbf{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{q\mathbf{v} \times \hat{\mathbf{r}}}{r^2}$$

where $\mathbf{r}$ points from source q to the point P

The universal constant $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$ is called the **magnetic constant** (has nothing to do with magnetic moment μ)

For a point charge moving with velocity $\mathbf{v}$, the magnetic field lines are circles centered on the line of $\mathbf{v}$ and lying in planes perpendicular to this line

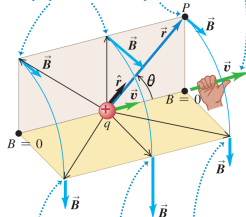
For a positive point charge moving with velocity $\mathbf{v}$, the magnetic field lines are found by the right hand rule: thumb in the direction of $\mathbf{v}$ and direction of rotation of other four fingers are the direction of the field lines

For a negative point charge use the same rule but with the left hand

(a) Perspective view

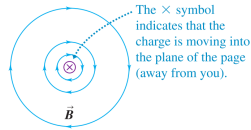
Right-hand rule for the magnetic field due to a positive charge moving at constant velocity:
Point the thumb of your right hand in the direction of the velocity. Your fingers now curl around the charge in the direction of the magnetic field lines. (If the charge is negative, the field lines are in the opposite direction.)

For these field points, $\vec{r}$ and $\vec{v}$ both lie in the beige plane, and $\vec{B}$ is perpendicular to this plane.



For these field points, $\vec{r}$ and $\vec{v}$ both lie in the gold plane, and $\vec{B}$ is perpendicular to this plane.

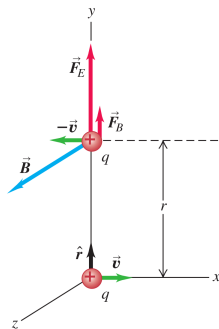
(b) View from behind the charge



Exercise 2

<https://epoll.srv.ualberta.ca> – Code: MBD

Two protons move parallel to the x -axis in opposite directions at the same speed v (small compared to the speed of light c). At the instant shown, find the electric and magnetic force vectors on the upper proton and compare their magnitudes. Note that the speed of light is $c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$.



Exercise 2

<https://epoll.srv.ualberta.ca> – Code: MBD

